DPUSCNX

This DPU represents the database SCNX.

It provides information about the road network at the position of the EGO vehicle.

Note: The coordinate systems used for the DPU variables are described in the section Coordinate Systems. Unless specified otherwise, the variables use the SILAB world coordinate system.

1. Inputs:

X	x-coordinate of the EGO vehicle. Unit: m
Y	y-coordinate of the EGO vehicle. Unit: m
Z	z-coordinate of the EGO-vehicle. Unit: m
yaw	yaw angle of the EGO-vehicle. Unit: rad
pitch	pitch angle of the EGO-vehicle. Unit: rad
roll	roll angle of the EGO-vehicle. Unit: rad
TLCThreshold	If the TLC current time to line crossing is below this threshold the output TLCWarning is set to 1. Unit: s, Default: 1.0 s
CondVar[1...20]	Inputs that are used for defining conditional connections of road sections.

2. Parameters:

Map	Name of the map that should be used during simulation. In the configuration section of the DPU a string is assigned to this parameter. See “Functional principle” for an example. Default: „Map1“
SetupPoint	Name of the setup point within the selected map at which the driver will start the simulation. In the configuration section of the DPU a string is assigned to this parameter. See “Functional principle” for an example. Default: „SetupPoint1“.
NLaneChangesLeftOnStart	Name of lane changes to the left that the EGO vehicle should perform at the start of the simulation. Default: 0
VisibleRoadLength	Visibility range of the driver along the road. Unit: m, Default: 2000 m.
VehicleWidth	Width of the EGO-vehicle, i.e. the wheelbase along the Y axis. Unit: m, Default: 1.665 m
VehicleLength	Length of the EGO-vehicle, i.e. the wheelbase along the X axis. Unit: m, Default: 3.9 m

SuperStructureManagedCursorID	ID of the managed cursor around which superstructure objects will be placed dynamically (e.g. light posts). Default: 0, i.e. the primary cursor (EGO-vehicle)
IgnoreOutOfMap	Default: 0, which means that the vehicle will stay at the road height after driving over the end of a road. If the value is set to 1, the car will "jump down" to the landscape height.
EnableRefinement	Default: 0, which means that bumps on the road are ignored. If this value is set to 1, the used road texture, as well as object strips and some objects (such as storm drains) will lead to bumps that can be seen in the ZOut, PitchRoad, and mu outputs.

3. Outputs:

AEditX *AEditY*	In Area2 sections: Current position in m measured in the SILABAEdit coordinate system. In course segments always 0. Unit: m
psi	Difference between yaw angle and tangent angle of the current lane. Positive values signify a deviation to the right. If the driver drives exactly against the direction of the lane this value is π. Unit: rad; range: [-π, π]
psi_OriInd	Difference between yaw angle and tangent angle of the current lane, taking into account the driving direction. (If the driver uses the lane against its direction (e.g. while overtaking), π is added to the value.) Positive values signify a deviation to the right. If the driver drives exactly against the direction of the lane this value is 0. Unit: rad; range: [-2π, 2π]
TangentAngleXY	Tangent angle of the current lane in the world coordinate system. The value increases along lanes that turn to the right. Unit: rad; range: [-π, π]
TangentAngleXY_OriInd	Tangent angle of the current lane in the world coordinate system, taking into account the driving direction. (If the driver uses the lane against its direction (e.g. while overtaking), π is added to the value.) Unit: rad; range: [-2π, 2π]
s	On Course sections: Position along the road. In Area2 sections this value is always 0. Einheit: m
IsOnTargetRoute	In an Area2 and if in this Area2 a target route is defined: Is the driver on this target route?
sTargetRoute	In an Area2 and if in this Area2 a target route is defined: Position along this target route. On Course sections or if in an Area2 no target route is defined: 0. Unit: m
sPropagated	On Course sections: Length of the road since start of the simulation. In Area2 sections this value always is that of entering the Area2. Unit: m
sLane	Position along the current lane. Unit: m

sCombined	On Course sections: s, on Area2 sections: sTargetRoute. For a description of s and sTargetRoute see above. Unit: m
LaneCurvatureXY	Lane curvature in the XY plane of the world coordinate system. Positive values signify a curve to the right. Example: Within a circular curve with radius 1000 m, the curvature is 0.001. Unit: 1/m
LaneGrade	Road gradient in the SZ plane (i.e. along the road). Positive values are upwards. Unit: %
LaneCurvatureZ	Lane curvature in the SZ-plane. Positive values signify that the LaneGrade is increasing at the current position. Example: Within a section with fillet radius 1000 m, the curvature is 0.001. Unit: 1/m
LaneIdx	Index of the current lane. On Course sections the lanes are numbered from left to right starting with 0 (according to the cross section profile). In Area2 sections the lane numbering is generated by SILABAEdit. For example, if SILABAEdit shows „l13“ for a lane then LaneIdx = 13.
LaneLength	Length of the current lane. In Course sections this is the length of the current straight or bend segment. In Area2 sections this is the length of the lanes as defined in SILABAEdit. Unit: m
LaneWidth	Width of the current lane. Unit: m
LateralDistance	Distance of the center of gravity of the EGO-vehicle to the middle of the current lane (ideal lane). Negative distances are left of the ideal lane. Unit: m
LateralPos	In Course sections: Distance of the center of gravity of the EGO-vehicle from the right border of the road. Positive distances are right of the right road border. In Area2 sections this value is not defined. Unit: m
LaneGapLeft	Gap between left side of the EGO-vehicle and left border of the current lane (seen in the direction of travel). Unit: m
LaneGapRight	Gap between right side of the EGO-vehicle and right border of the current lane (seen in the direction of travel). Unit: m
vLateral	Velocity with wich the EGO-vehicle approaches the lane border, measured perpendicular to the vehicles longitudinal axis. Unit: m/s
NLanes	In Course sections: Total number of roads in the current cross section profile. In Area2 sections this value is always 1.
TLC	In Course sections: Time to Lane Crossing with respect to both borders of the lane. More precisely, the time until one of the vehicle’s

	wheels (whose positions are defined by VehicleWidth and VehicleLength) leaves the lane. In Area2 sections, this value is undefined. Unit: s
TLCLeft	In Course sections: Time to Lane Crossing with respect to the left border of the lane (seen in the direction of travel). More precisely, the time until one of the vehicle's wheels (whose positions are defined by VehicleWidth and VehicleLength) leaves the lane on the left side. If psi > 0, this value is 0. In Area2 sections this value is undefined. Unit: s
TLCRight	In Course sections: Time to Lane Crossing with respect to the right border of the lane (seen in the direction of travel). In Area2 sections this value is undefined. Unit: s
TLCWarning	In Course sections: Warning (=1) if TLC is below TLCThreshold. In Area2 sections this value is undefined.
OBSLaneIdx	Additional lane index. An OBSLaneIdx can be defined for one or multiple lanes in SILABAEdit (for Area2 sections) or in the script (for Course sections).
Orientation	Orientation of the EGO-vehicle on the current lane. The value 0 means that the driver travels along the direction of the lane, the value 1 means that the driver travels against the direction of the lane.
XOffset, YOffset, ZOffset	These outputs are used during the simulation to move the vehicle in addition to the vehicle dynamics, e.g. to keep it on the road in bridge or tunnel sections. This output normally serves as input for vehicle dynamics DPUs. Unit: m
SetupDone	Defines if the vehicle setup is completed (1) or still running (0). The Setup variables defined below are only valid if SetupDone =1. The output is only relevant during SILAB's launch phase; it will always be 1 while the simulation is running.
SetupX, SetupY, SetupZ	Position of the setup point with respect to the origin of the world coordinate system. This output normally serves as input for vehicle dynamics DPUs. The output is static and available during SILAB's launch phase. Unit: m
SetupYaw, SetupPitch, SetupRoll	Rotation of the setup point with respect to the origin of the world coordinate system. This output normally serves as input for vehicle dynamics DPUs. Unit: rad
PitchRoad	Pitch angle of the ground surface (road or landscape). Unit: rad
RollRoad	Roll angle of the ground surface (road or landscape). Unit: rad
ModuleTemplateID	Module type ID, as given in the road network definition.
ModuleInstanceID	Number of the instance of the current module type. The instances are counted in the order as they are defined in the script.

ModuleInstanceCounter	Counter that is incremented each time the driver enters a module instance (independent of the module type).
ModuleTemplateName	Module type name, as given in the road network definition (or as specified in SILABAEdit).
ModuleInstanceName	Name of the module instance, as given in the road network definition.
NodeTemplateID	Course/Area2 type ID, as given in the road network definition.
NodeInstanceID	Number of the instance of the current Course/Area2 type. The instances are counted in the order as they are defined in the script.
NodeInstanceCounter	Counter that is incremented each time the driver enters a Course/Area2 instance (independent of the Course/Area2 type).
NodeTemplateName	Type name of the Course or Area2, as given in the road network definition (or as specified in SILABAEdit).
ModuleInstanceName	Name of the Course or Area2 instance, as given in the road network definition (or as specified in SILABAEdit).
CurrentNodeIsArea2	1, if the EGO-vehicle is within an Area2 section, 0 otherwise.
CurrentNodeIsCourse	1, if the EGO-vehicle is within a Course section, 0 otherwise.
mu	Coefficient of friction. For Course sections this value is defined in the cross section profile. For Area2 sections this value is defined in SILABAEdit using the tool for additional lane information.
GPS_N, GPS_E, GPS_Heading_deg	GPS coordinates of the EGO vehicle. In Area2 sections, this requires that you have specified the GPS data for the model image on which the Area2 is based in SILABAEdit. For Course sections, GPS coordinates can be specified in the configuration file.

4. Remarks to the outputs:

If the driver is in an Area2 section and if in this Area2 a target route is defined then outputs for lane information refer to the lane of the target route that is closest to the driver. This concerns the following outputs:

psi, *psi_OriInd*, *TangentAngleXY*, *TangentAngleXY_OriInd*, *LaneCurvatureXY*, *LaneIdx*, *LaneLength*, *LaneWidth*, *LateralDistance*, *vLateral*, *Orientation*.

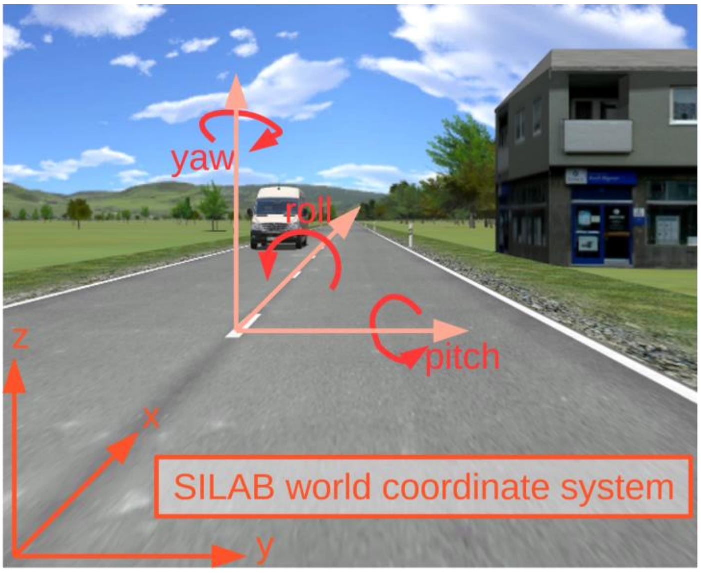
The definition of a target route is described in the document „SILABAEdit“ (layer „lanes“).

5. Launch data:

SCN_SetupPoint	Name of the setup point (in the current map) at which the drive will be placed at start of the simulation.
SCN_CondVar1 … SCN_CondVar10	Initial values for the inputs CondVar1 up to CondVar10

6. Coordinate systems:

Most inputs and outputs of DPUSCNX use SILAB’s world coordinate system, which is defined as follows:



- The driver starts near the origin of the coordinate system, with the vehicle pointing towards the positive **X** axis.
- Positive **Y** values are to the right
- Positive **Z** values are to the top
- **yaw** is a rotation about the Z axis, positive angles are to the right
- **pitch** is a rotation about the Y axis, positive angles are downwards
- **roll** is a rotation about the X axis, positive angles are an inclination to the left

The outputs AEditX and AEditY use the *SILABAEdit* coordinate system, which is shown in the rulers when the Area2 file is opened in *SILABAEdit*:

- Positive **X** values are to the right
- Positive **Y** values are upwards

7. Functional principle:

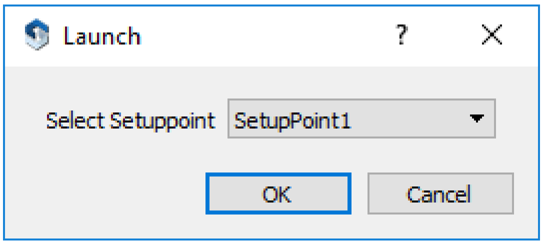
The DPUSCNX represents the database SCNX of SILAB. During preparation of the simulation, the map with the name given in *Map* is read from the configuration script . At simulation start the driver is placed at the setup point with the name given in *SetupPoint*. The parameters Map and *SetupPoint* are initialized in the configuration section of the DPUSCNX.

8. Example:

```
DPUSCNX scn
{
      Computer = { ... };
```

```
      Index = ...;
      Map = "MyMap";
      SetupPoint = "MySetupPoint";
};
```

Alternatively, the setup point can be chosen by the user. Therefore, the launch data „SCN_SetupPoint“ is used. The following combo box appears in the launch step of SILAB:



To do so, the following has to be included in the SILAB System section of the configuration script:

```
SILAB System
{
      LaunchData1 = ("Select Setuppoint", SCN_SetupPoint,
                        LD_COMBO_DEFAULT, "SetupPoint1");
};
```

Please note that for this DPUSCNX has to be executed on the operator computer. If this causes performance problems (Updating the database during simulation uses quite an amount of processing power which might not be available on the operator computer.) the DPUSCNXLaunch can be used as an alternative.

During simulation the DPUSCNX receives the coordinates as well as yaw, pitch and roll angles of the EGO-vehicle in each simulation step (normally from the vehicle dynamics DPU). DPUSCNX searches the point in the road network that is closes to the EGO-vehicle. It then updates the road network and the surrounding landscape within the drivers range of visiblity. Afterwards the outputs of the DPU are updated.

The following information is only of interest if the standard configuration architecture (included in the scope of delivery of SILAB) is **not** used as basis for the DPU configuration.

In the DPU configuration the DPUSCNX should be computed after the vehicle dynamics DPU by selecting the parameter Index accordingly (see „ Reference SILAB Configuration and Operation“). The DPU wiring has the following structure:

The distances *SetupX*, *SetupY* and *SetupZ*, and the angles *SetupYaw*, *SetupPitch* and *SetupRoll* are computed once during the launch phase of SILAB. These values define the location of the setup point in the world coordinate system. The vehicle dynamics should be initialized at this location and orientation.

The values *XOffset*, *YOffset* and *ZOffset* are computed while the simulation is running. once in the first simulation step. In the vehicle dynamics DPU these values have to be added to the initial coordinates of the vehicle dynamics simulation. By doing so, the driver is placed on the rightmost lane of setup road.